

GOVERNMENT COLLEGE OF ENGINEERING (IRTT)

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Electronics and Communication Engineering

SMART IRRIGATION SYSTEM

A project report Submitted by

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ABSTRACT

A smart irrigation system is an advanced solution designed to optimize water usage in agriculture and landscaping by leveraging modern technology. This system integrates sensors, real-time weather data, and automation to monitor soil moisture levels, temperature, humidity, and precipitation. Based on this data, the system dynamically adjusts the watering schedule to meet the specific needs of plants, reducing water waste and improving crop yield. By utilizing IoT (Internet of Things) technology, smart irrigation systems provide remote control and monitoring capabilities via mobile apps or web interfaces, allowing for more efficient water management and conservation. This system not only promotes sustainable agriculture but also reduces operational costs and enhances productivity, making it a vital tool for modern farming and landscaping.

INTRODUCTION

A smart irrigation system is an innovative technological advancement designed to improve water management in agriculture and landscaping. Traditional irrigation systems often result in overwatering or underwatering due to a lack of real-time environmental feedback, leading to inefficient water use, higher costs, and potential damage to crops or landscapes. Smart irrigation systems address these challenges by utilizing sensors and real-time data, such as soil moisture, temperature, humidity, and weather forecasts, to automatically adjust water delivery based on the specific needs of plants.

These systems are typically connected to the Internet of Things (IoT), allowing users to monitor and control irrigation schedules remotely via smartphones or computers. By optimizing water usage, smart irrigation systems not only conserve water but also enhance crop yields, reduce labor costs, and minimize environmental impact. With the increasing need for sustainable agricultural practices and resource conservation, smart irrigation systems offer a practical and efficient solution to ensure water is used effectively and responsibly.

PROBLEM STATEMENT

Since, the water is irrigated directly in the land, plants under go high stress from variation in soil moisture, therefore plant appearance is reduced. The absence of automatic controlling of the system result in improper water control system.

PROBLEM SOLUTION

A smart irrigation system uses sensors to monitor soil moisture, temperature, humidity, and weather conditions in real-time.

It automates water delivery through drip or sprinkler systems using a microcontroller connected to IoT for remote control.

A mobile app allows users to set irrigation schedules, view data, and receive alerts.

The system optimizes water usage by only irrigating when needed, reducing water waste and improving crop yields.

It can also integrate machine learning for predictive irrigation and can be powered by solar energy for sustainable, off-grid operation.

The system is scalable and adaptable for different farm sizes and crops.

WORKING PRINCIPLE

A smart irrigation system works by automating and optimizing the watering process based on real-time data, ensuring efficient water use and healthier plant growth. The key components of a smart irrigation system and their working principles are as follows:

1. Sensors

-Soil Moisture Sensors: These sensors monitor the moisture content of the soil in real-time. When the soil moisture drops below a set threshold, the system triggers irrigation.

Weather Sensors: Some systems use weather data (temperature, humidity, rainfall) to predict when and how much water is needed. If rain is predicted, the system can delay watering to avoid over-irrigation.

Temperature Sensors: High temperatures may increase evaporation and plant water needs, prompting the system to adjust irrigation.

2.Control Unit (Controller)

- The controller acts as the brain of the system. It processes data from sensors and external sources (e.g., weather forecasts) to determine the watering schedule.

- Many smart controllers are Wi-Fi or Bluetooth-enabled, allowing users to remotely manage the system through a smartphone app or web interface.

- Users can set specific parameters, such as watering duration, soil type, and plant needs, which the system uses to fine-tune watering cycles.

3. Automated Valves and Sprinklers

- The control unit activates valves connected to irrigation pipes and sprinklers. It opens or closes the valves to allow water flow based on the data from the sensors and programmed schedules.

- Drip irrigation or sprinklers deliver water directly to the plants, minimizing water waste through runoff or evaporation.

4. Cloud Integration & Data Analytics

- Some smart irrigation systems are connected to cloud platforms where they can receive weather updates and store historical data on water usage and plant growth.

- Advanced systems may use artificial intelligence or machine learning algorithms to improve watering efficiency over time, adjusting schedules based on trends and seasonal patterns.

5. User Interface

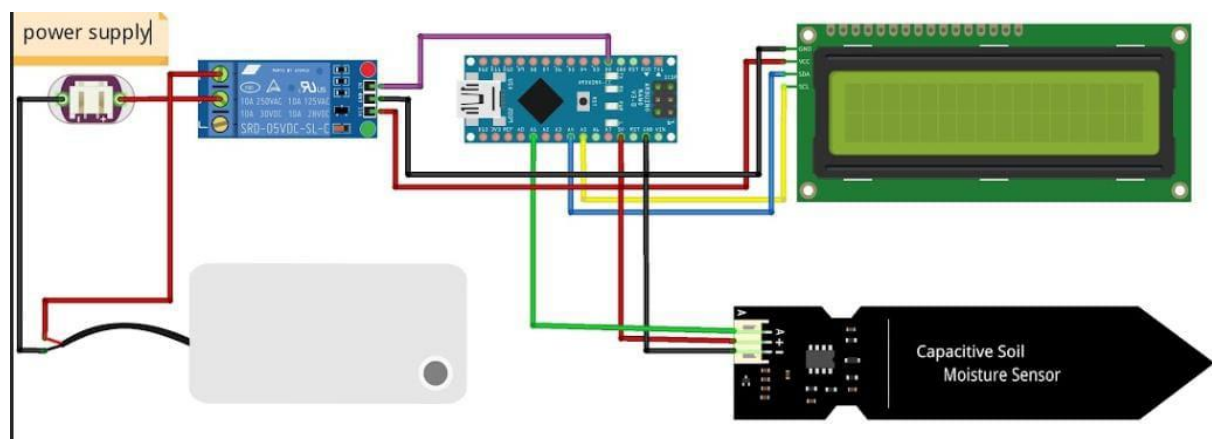
- Through mobile apps or web dashboards, users can monitor the system, check soil moisture levels, and modify settings. They can also receive alerts if there are issues like water leakage or abnormal moisture levels.

Working Process:

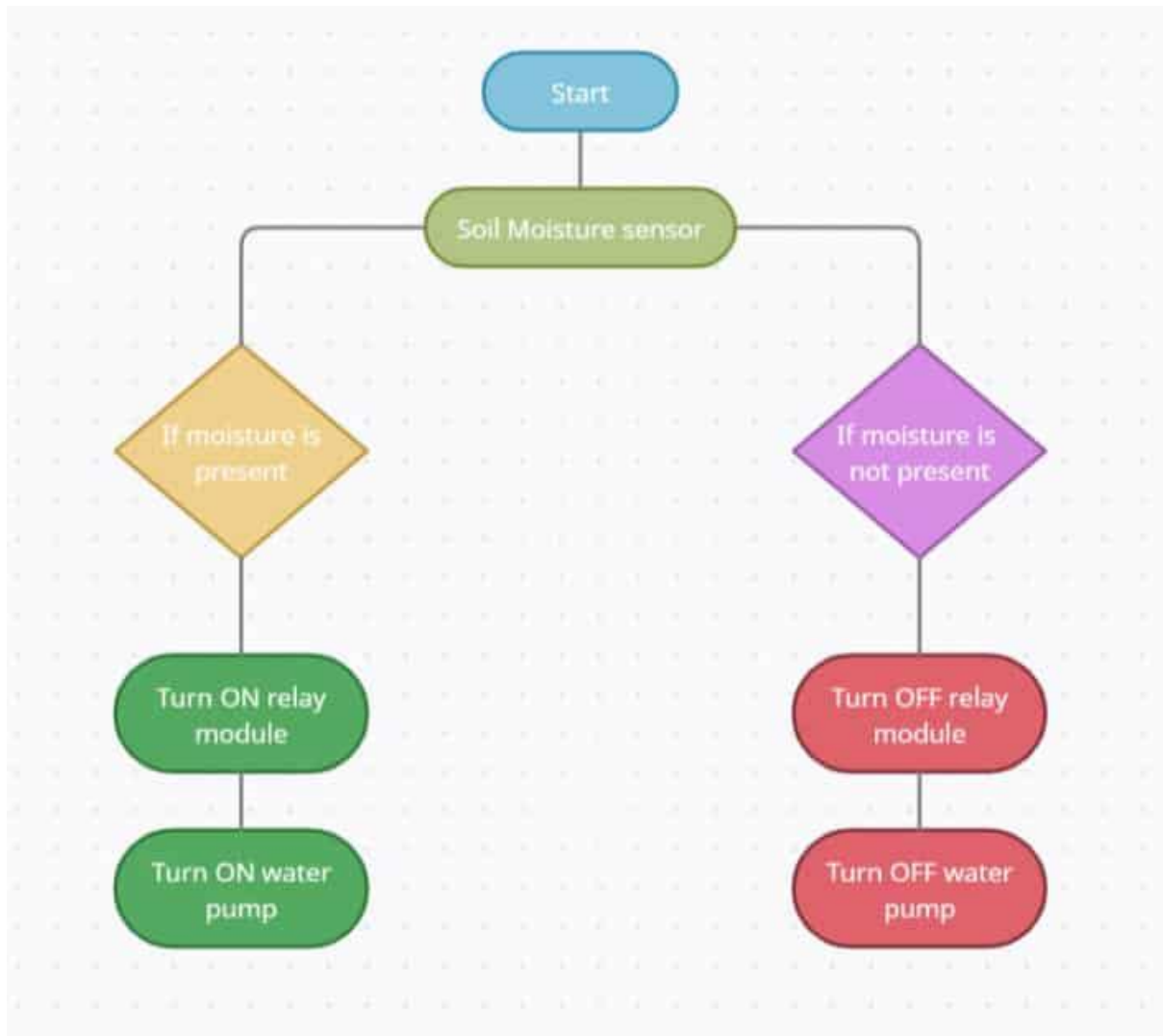
1. **Data Collection:** The sensors gather real-time environmental data such as soil moisture, temperature, and rainfall.
2. **Decision Making:** The controller processes this data and, if necessary, adjusts the irrigation schedule based on the plant's needs and environmental conditions.
3. **Action:** The system activates the irrigation when required, supplying the right amount of water.
4. **Monitoring:** The system continues to collect data and make adjustments, ensuring water is used efficiently.

This combination of real-time data and automation helps conserve water, reduce costs, and support sustainable agriculture or landscaping.

CIRCUIT DIAGRAM



SEQUENTIAL DIAGRAM



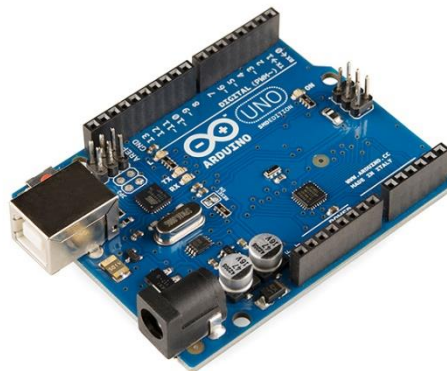
COMPONENTS REQUIRED

UNO ARDIUNO

The Arduino UNO is a standard board of Arduino. Here UNO means 'one' in Italian. It was named as UNO to label the first release of Arduino Software. It was also the first USB board released by Arduino. It is considered as the powerful board used in various projects. Arduino.cc developed the Arduino UNO board.

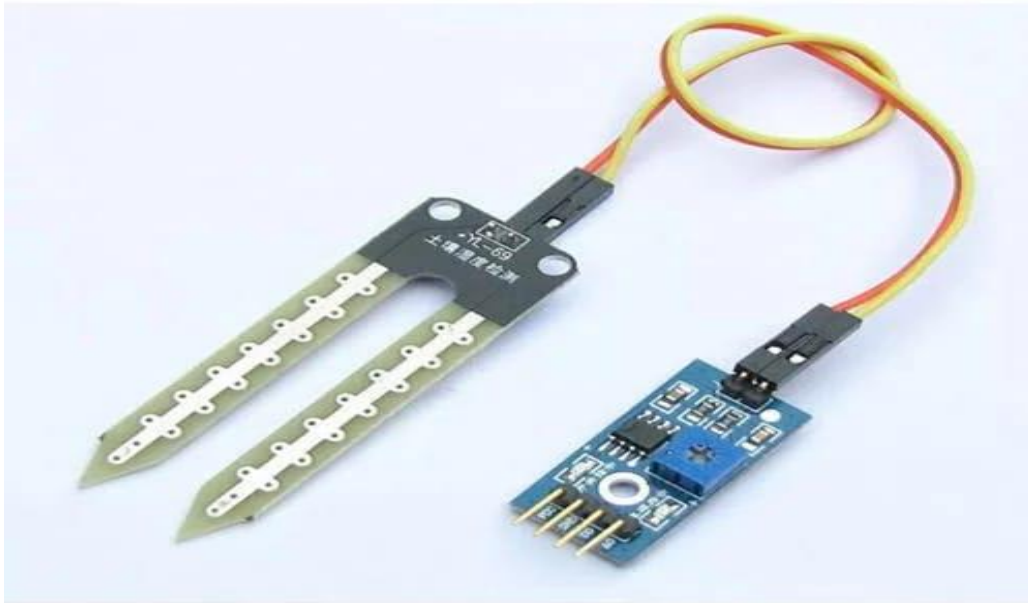
Arduino UNO is based on an ATmega328P microcontroller. It is easy to use compared to other boards, such as the Arduino Mega board, etc. The board consists of digital and analog Input/Output pins (I/O), shields, and other circuits.

The Arduino UNO includes 6 analog pin inputs, 14 digital pins, a USB connector, a power jack, and an ICSP (In-Circuit Serial Programming) header. It is programmed based on IDE, which stands for Integrated Development Environment. It can run on both online and offline platforms.



MOISTURE SENSOR

Soil moisture sensors measure the volumetric water content in soil.[1] Since the direct gravimetric measurement of free soil moisture requires removing, drying, and weighing of a sample, soil moisture sensors measure the volumetric water content indirectly by using some other property of the soil, such as electrical resistance, dielectric constant, or interaction with neutrons, as a proxy for the moisture content.



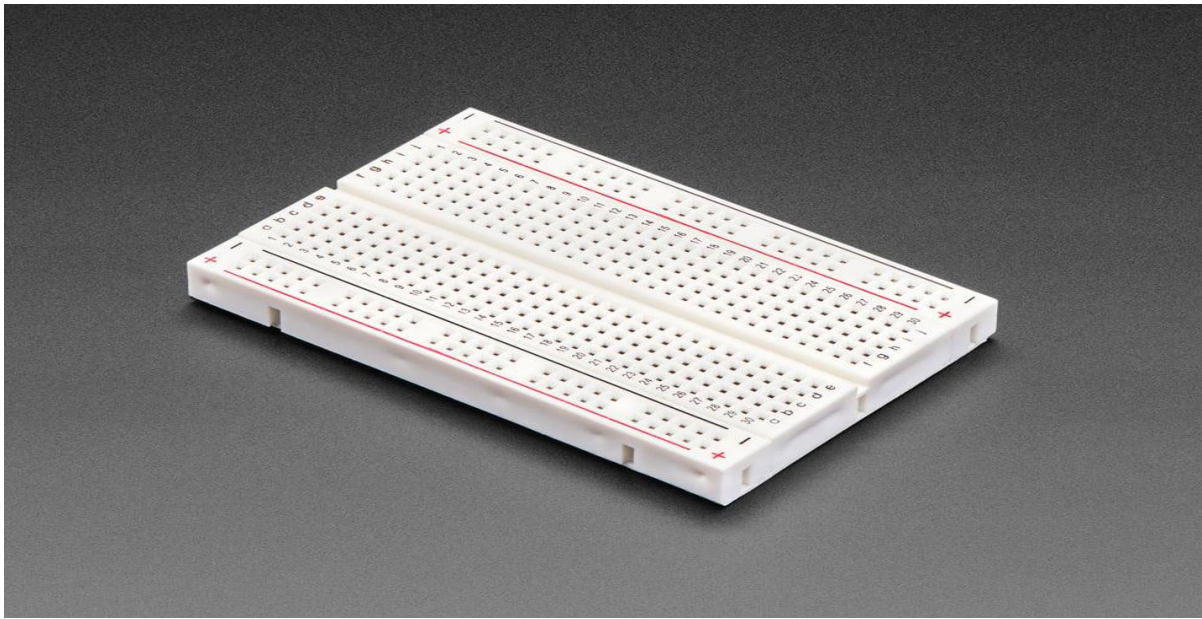
BREADBOARD

A breadboard is a tool used to build and test electronic circuits without soldering. It consists of a plastic board with a grid of holes, designed to hold components like resistors, capacitors, and integrated circuits temporarily. The holes are connected by metal strips underneath, allowing for easy insertion and removal of wires and components.

Breadboards are commonly used in prototyping because they are reusable and versatile, enabling quick modifications to circuits. There are two main sections: the terminal strips, where most components are placed, and the power rails, which distribute power across the circuit.

Key features:

- No soldering required, making it ideal for beginners and temporary projects.
- Components can be easily rearranged and reused.
- The standard layout consists of rows and columns with connections underneath to simplify wiring.



JUMPER WIRE

Jumper wires are insulated electrical wires used to connect components on a breadboard or between different devices in a circuit. They are essential for prototyping and testing circuits, allowing users to create connections without soldering.

Jumper wires come in different types, primarily:

- Male-to-male: Both ends have metal pins, ideal for connecting breadboards or components with female headers.
- female-to-female*: Both ends have sockets, used to connect pins on components like sensors or modules.
- male-to-female: One end has a pin and the other a socket, useful for connecting male and female pins.

Key features:

- Available in various lengths and colors for easy identification.
- Reusable and ideal for temporary connections.



5V DC WATER PUMP

A 5V DC water pump is a small, low-power pump that operates on a 5-volt direct current (DC) power supply. It is commonly used in electronics projects, robotics, and small-scale water circulation systems such as fountains, cooling setups, or DIY irrigation systems. The pump typically has a compact design, making it suitable for space-constrained applications.

Key features:

- Low voltage: Operates on 5V DC, making it compatible with USB power sources or microcontroller boards like Arduino.
- Submersible or inline: Some pumps are designed to be fully submerged in water, while others work inline to push water through tubing.
- Low power consumption: Efficient for small applications, drawing minimal **current**.
- Flow rate: Typically, the flow rate is low to moderate, suitable for hobby projects and small fluid transfers.



9V BATTERY

A 9V battery is a rectangular-shaped, commonly used power source for a wide variety of electronic devices. It typically delivers 9 volts of direct current (DC) and consists of six 1.5V cells arranged in series. Known for its distinctive shape with snap terminals on top, the 9V battery is used in applications like smoke detectors, portable devices, toys, and some electronic projects.

Key features:

- Voltage: Provides 9 volts of power, ideal for devices requiring moderate voltage.
- Size and shape: Compact rectangular form, making it easy to use in small devices.

- Capacity: Typically ranges from 500mAh to 1200mAh, depending on the type (e.g., alkaline, lithium).
- Common types: Includes alkaline, lithium, and rechargeable versions, with different capacities and lifespans.
- Applications: Frequently used in household electronics, small appliances, and hobby projects.



PROGRAM

```
int soil=A0;  
int relay=3;  
  
void setup() {  
  
  pinMode(soil,INPUT);  
  pinMode(relay,OUTPUT);  
  digitalWrite(relay,HIGH);  
  Serial.begin(9600);  
}
```

```
void loop() {  
  
    int soilData=analogRead(soil);  
    Serial.print("Soil DATA:");  
    Serial.println(soilData);  
  
    if(soilData>900)  
    {  
        digitalWrite(relay,LOW);  
    }  
    else  
    {  
        digitalWrite(relay,HIGH);  
    }  
}
```



CONCLUSION

In conclusion, a smart irrigation system represents a significant advancement in sustainable agriculture and water management. By leveraging technology such as sensors, weather data, and automated controls, these systems optimize water usage, reduce waste, and enhance crop yields. They not only contribute to environmental conservation but also offer economic benefits for farmers by lowering operational costs and increasing efficiency. As water scarcity becomes a growing concern, the adoption of smart irrigation solutions will be crucial in promoting responsible water use and ensuring food security for the future. Embracing this technology can lead to a more sustainable and resilient agricultural landscape.

